CURRICULUM BOOK

UNDERGRADUATE PROGRAM: BACHELOR OF ENGINEERING

COMPUTER ENGINEERING

Effective from Academic Year: 2026-27

Status: Autonomous College Syllabus Handbook

Dr. Deepak Bhoir	**Dr. Sujata P. Deshmukh**	**Dr. Surendra Rathod**
Dean Academics	HOD (COMP)	Principal

ACADEMIC PREAMBLE

As an autonomous engineering college affiliated with the University of Mumbai, Fr. Conceicao Rodrigues College of Engineering is dedicated to engineering excellent learners, innovators, and future technical leaders. Autonomy enables the institution to respond to technological shifts and design responsive curricula that balance theoretical foundations with experiential design, modern tools, and professional ethics.

This curriculum structure represents a unified framework curated in collaboration with academic advisors and leading technical firms. The curriculum emphasizes robust mathematical models, project-based validation, and outcome-oriented professional practice.

Vision of the Department

To be a leading center of excellence in engineering education and research, fostering industry-ready technical innovators, ethically sound professionals, and empathetic citizens.

Mission of the Department

- Deliver standard-aligned engineering education using student-centric pedagogy and laboratory environments.
- Foster research, technical incubation, and industry-institution partnerships.
- Cultivate life-long learning habits, teamwork capabilities, and professional ethical standards.

CURRICULUM ARCHITECTURE AND NEP ALIGNMENT

Course Nomenclature Table

Category	Course Category Name	NEP-2020 Credit Framework Description
BS	Basic Sciences	Fundamental disciplines including Physics, Chemistry, and Advanced Calculus.
ES	Engineering Sciences	Interdisciplinary foundations such as Computing, Graphics, and Electrical Systems.
PC	Professional Core	Core discipline-specific subjects, advanced concepts, and matching practicals.
PE	Professional Electives	Branching streams for specialized expertise such as AI/ML, Security, and IoT.
OE	Open Electives	Multi-disciplinary courses offered across departments for wider learning.
HS	Humanities and Social Sciences	Professional Communication, Business Ethics, IPR, and Management.

SEMESTERWISE CURRICULUM STRUCTURE

Computer Engineering Program: Semester 3

Course Code	Course Name	Teaching Scheme (Contact Hours/Week)			Credits Assigned				Examination Scheme						
									Theory Courses				Practical / Lab		Total Marks
		L	T	P	L	T	P	Total	ISE 1	ISE 2	MSE	ESE	ISE 1	ISE 2	
CSC301	**Data Structures and Algorithms**	3	1	0	3	1	0	**4**	20	20	-	60	-	-	**100**
CSL301	**Data Structures Laboratory**	0	0	2	0	0	1	**1**	20	20	-	-	25	25	**100**
CSC302	**Database Management Systems**	3	0	0	3	0	0	**3**	20	20	-	60	-	-	**100**
CSL302	**Database Systems Laboratory**	0	0	2	0	0	1	**1**	20	20	-	-	25	25	**100**
CSC303	**Computer Networks**	3	1	0	3	1	0	**4**	20	20	-	60	-	-	**100**
CSC304	**Operating Systems**	3	0	0	3	0	0	**3**	20	20	-	60	-	-	**100**
CSE305	**Professional Elective: Artificial Intelligence**	3	0	0	3	0	0	**3**	20	20	-	60	-	-	**100**
CSP306	**Mini Project I**	0	0	4	0	0	2	**2**	20	20	-	-	25	25	**100**
Total		15	2	8	**21**				-						**800**

Abbreviations: L: Lecture; T: Tutorial; P: Practical/Lab; ISE: In-Semester Evaluation; MSE: Mid-Semester Examination; ESE: End Semester Examination.

Regulatory Ordinance Code: R-2026-COMP-III

SEMESTERWISE CURRICULUM STRUCTURE

Computer Engineering Program: Semester 4

Course Code	Course Name	Teaching Scheme (Contact Hours/Week)			Credits Assigned				Examination Scheme						
									Theory Courses				Practical / Lab		Total Marks
		L	T	P	L	T	P	Total	ISE 1	ISE 2	MSE	ESE	ISE 1	ISE 2	
CSC401	**Design and Analysis of Algorithms**	3	1	0	3	1	0	**4**	20	20	-	60	-	-	**100**
CSL401	**Algorithm Design Laboratory**	0	0	2	0	0	1	**1**	20	20	-	-	25	25	**100**
CSC402	**Software Engineering**	3	0	0	3	0	0	**3**	20	20	-	60	-	-	**100**
CSC403	**Computer Architecture**	3	1	0	3	1	0	**4**	20	20	-	60	-	-	**100**
CSE404	**Professional Elective: Cyber Security**	3	0	0	3	0	0	**3**	20	20	-	60	-	-	**100**
CSO405	**Open Elective: Data Visualization**	2	0	2	2	0	1	**3**	20	20	-	60	25	25	**150**
CSP406	**Community Innovation Project**	0	0	4	0	0	2	**2**	20	20	-	-	25	25	**100**
Total		14	2	8	**20**				-						**750**

Abbreviations: L: Lecture; T: Tutorial; P: Practical/Lab; ISE: In-Semester Evaluation; MSE: Mid-Semester Examination; ESE: End Semester Examination.

Regulatory Ordinance Code: R-2026-COMP-IV

TABLE OF CONTENTS

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSC301	Data Structures and Algorithms	3	1	--	--	3	1	--	4

Examination Scheme						
	ISE1	MSE	ISE2	ESE	Total	
Theory	20	--	20	60	100	

Pre-requisite Course Codes		Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design data structures and algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**1.1**	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Module No.	Unit No.	Topics	Ref.	Hrs.
2	**2.1**	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	**3.1**	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	**4.1**	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	**5.1**	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Data Structures and Algorithms: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Data Structures and Algorithms. Pearson, 1st, 2024.
Department Faculty Board. Data Structures and Algorithms Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content (Practical Only)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSL301	Data Structures Laboratory	--	--	2	--	--	--	1	1

Examination Scheme						
	ISE1	MSE	ISE2	ESE	Total	
Theory	--	--	--	--	--	
Lab	25	--	25	--	50	

Pre-requisite Course Codes	Nil	
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design data structures laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Course Assessment: (Lab)

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics wise List of Experiments with relevant topic	**Ref.**	**COs**
1	Data Structures Laboratory Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Data Structures Laboratory Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Data Structures Laboratory Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Data Structures Laboratory Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Data Structures Laboratory Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Data Structures Laboratory Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Data Structures Laboratory: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Data Structures Laboratory. Pearson, 1st, 2024.
Department Faculty Board. Data Structures Laboratory Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSC302	Database Management Systems	3	--	--	--	3	--	--	3

Examination Scheme						
	ISE1	MSE	ISE2	ESE	Total	
Theory	20	--	20	60	100	

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.	
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design database management systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**1.1**	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Module No.	Unit No.	Topics	Ref.	Hrs.
2	**2.1**	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	**3.1**	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	**4.1**	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	**5.1**	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Database Management Systems: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Database Management Systems. Pearson, 1st, 2024.
Department Faculty Board. Database Management Systems Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content (Practical Only)

<table>
<tr><th rowspan="3">Course Code

CSL302</th><th rowspan="3">Course Name

Database Systems Laboratory</th><th colspan="4">Teaching Scheme (Hrs/week)</th><th colspan="4">Credits Assigned</th></tr>
<tr><th>L</th><th>T</th><th>P</th><th>SL</th><th>L</th><th>T</th><th>P</th><th>Total</th></tr>
<tr><td>--</td><td>--</td><td>2</td><td>--</td><td>--</td><td>--</td><td>1</td><td>1</td></tr>
</table>

Examination Scheme						
	ISE1	**MSE**	**ISE2**	**ESE**	**Total**	
Theory	--	--	--	--	--	
Lab	25	--	25	--	50	

<table>
<tr><td colspan="2">Pre-requisite Course Codes</td><td>Nil</td></tr>
<tr><td colspan="3">After the successful completion students should be able to:</td></tr>
<tr><td rowspan="5">Course Outcomes</td><td>CO1</td><td>Identify database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.</td></tr>
<tr><td>CO2</td><td>Explain database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.</td></tr>
<tr><td>CO3</td><td>Apply database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.</td></tr>
<tr><td>CO4</td><td>Analyze database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.</td></tr>
<tr><td>CO5</td><td>Design database systems laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.</td></tr>
</table>

Course Assessment: (Lab)

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics wise List of Experiments with relevant topic	**Ref.**	**COs**
1	Database Systems Laboratory Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Database Systems Laboratory Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Database Systems Laboratory Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Database Systems Laboratory Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Database Systems Laboratory Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Database Systems Laboratory Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Database Systems Laboratory: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Database Systems Laboratory. Pearson, 1st, 2024.
Department Faculty Board. Database Systems Laboratory Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSC303	Computer Networks	3	1	--	--	3	1	--	4

Examination Scheme						
	ISE1	MSE	ISE2	ESE	Total	
Theory	20	--	20	60	100	

Pre-requisite Course Codes		Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design computer networks concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**1.1**	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Module No.	Unit No.	Topics	Ref.	Hrs.
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Computer Networks: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Computer Networks. Pearson, 1st, 2024.
Department Faculty Board. Computer Networks Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

<table>
<tr><td rowspan="6">Course Code

CSC304</td><td rowspan="6">Course Name

Operating Systems</td><td colspan="4">Teaching Scheme (Hrs/week)</td><td colspan="4">Credits Assigned</td></tr>
<tr><td>L</td><td>T</td><td>P</td><td>SL</td><td>L</td><td>T</td><td>P</td><td>Total</td></tr>
<tr><td>3</td><td>--</td><td>--</td><td>--</td><td>3</td><td>--</td><td>--</td><td>3</td></tr>
</table>

Examination Scheme						
	ISE1	**MSE**	**ISE2**	**ESE**	**Total**	
Theory	20	--	20	60	100	

Pre-requisite Course Codes	Nil	
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design operating systems concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**1.1**	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Module No.	Unit No.	Topics	Ref.	Hrs.
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Operating Systems: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Operating Systems. Pearson, 1st, 2024.
Department Faculty Board. Operating Systems Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSE305	Professional Elective: Artificial Intelligence	3	--	--	--	3	--	--	3

Examination Scheme						
	ISE1	MSE	ISE2	ESE	Total	
Theory	20	--	20	60	100	

Pre-requisite Course Codes		Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design professional elective: artificial intelligence concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**1.1**	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Module No.	Unit No.	Topics	Ref.	Hrs.
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Professional Elective: Artificial Intelligence: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Professional Elective: Artificial Intelligence. Pearson, 1st, 2024.
Department Faculty Board. Professional Elective: Artificial Intelligence Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content (includes Practical)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSP306	Mini Project I	--	--	4	--	--	--	2	2

Examination Scheme						
	ISE1	MSE	ISE2	ESE	Total	
Theory	20	--	20	--	50	
Lab	25	--	25	--	50	

Pre-requisite Course Codes		Nil
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design mini project i concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**1.1**	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**0**	

Module No.	Unit No.	Topics	Ref.	Hrs.
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**0**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics	**Ref.**	**COs**
1	Mini Project I Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Mini Project I Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Mini Project I Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Mini Project I Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Mini Project I Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Mini Project I Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Mini Project I: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Mini Project I. Pearson, 1st, 2024.
Department Faculty Board. Mini Project I Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSC401	Design and Analysis of Algorithms	3	1	--	--	3	1	--	4

Examination Scheme						
	ISE1	MSE	ISE2	ESE	Total	
Theory	20	--	20	60	100	

Pre-requisite Course Codes		Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design design and analysis of algorithms concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**1.1**	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Module No.	Unit No.	Topics	Ref.	Hrs.
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Design and Analysis of Algorithms: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Design and Analysis of Algorithms. Pearson, 1st, 2024.
Department Faculty Board. Design and Analysis of Algorithms Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content (Practical Only)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	**T**	**P**	**SL**	**L**	**T**	**P**	**Total**
CSL401	**Algorithm Design Laboratory**	--	--	2	--	--	--	1	1

Examination Scheme						
	ISE1	**MSE**	**ISE2**	**ESE**	**Total**	
Theory	--	--	--	--	--	
Lab	25	--	25	--	50	

Pre-requisite Course Codes	Nil	
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design algorithm design laboratory concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Course Assessment: (Lab)

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics wise List of Experiments with relevant topic	**Ref.**	**COs**
1	Algorithm Design Laboratory Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Algorithm Design Laboratory Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Algorithm Design Laboratory Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Algorithm Design Laboratory Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Algorithm Design Laboratory Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Algorithm Design Laboratory Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Algorithm Design Laboratory: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Algorithm Design Laboratory. Pearson, 1st, 2024.
Department Faculty Board. Algorithm Design Laboratory Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSC402	Software Engineering	3	--	--	--	3	--	--	3

Examination Scheme						
	ISE1	MSE	ISE2	ESE	Total	
Theory	20	--	20	60	100	

Pre-requisite Course Codes		Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design software engineering concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**1.1**	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Module No.	Unit No.	Topics	Ref.	Hrs.
2	**2.1**	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	**3.1**	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	**4.1**	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	**5.1**	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Software Engineering: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Software Engineering. Pearson, 1st, 2024.
Department Faculty Board. Software Engineering Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSC403	Computer Architecture	3	1	--	--	3	1	--	4

Examination Scheme						
	ISE1	MSE	ISE2	ESE	Total	
Theory	20	--	20	60	100	

Pre-requisite Course Codes		Nil
After the successful completion students should be able to:		
Course Outcomes	CO1	Identify computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design computer architecture concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Module No.	Unit No.	Topics	Ref.	Hrs.
2	**2.1**	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	**3.1**	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	**4.1**	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	**5.1**	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Computer Architecture: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Computer Architecture. Pearson, 1st, 2024.
Department Faculty Board. Computer Architecture Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content

<table>
<tr><td rowspan="5">Course Code

CSE404</td><td rowspan="5">Course Name

Professional Elective: Cyber Security</td><th colspan="4">Teaching Scheme (Hrs/week)</th><th colspan="4">Credits Assigned</th></tr>
<tr><th>L</th><th>T</th><th>P</th><th>SL</th><th>L</th><th>T</th><th>P</th><th>Total</th></tr>
<tr><td>3</td><td>--</td><td>--</td><td>--</td><td>3</td><td>--</td><td>--</td><td>3</td></tr>
<tr><td colspan="8">
<table>
<tr><th colspan="7">Examination Scheme</th></tr>
<tr><th></th><th>ISE1</th><th>MSE</th><th>ISE2</th><th>ESE</th><th>Total</th><th></th></tr>
<tr><td>Theory</td><td>20</td><td>--</td><td>20</td><td>60</td><td>100</td><td></td></tr>
</table>
</td></tr>
</table>

<table>
<tr><td>Pre-requisite Course Codes</td><td colspan="2">Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.</td></tr>
<tr><td colspan="3">After the successful completion students should be able to:</td></tr>
<tr><td rowspan="5">Course Outcomes</td><td>CO1</td><td>Identify professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.</td></tr>
<tr><td>CO2</td><td>Explain professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.</td></tr>
<tr><td>CO3</td><td>Apply professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.</td></tr>
<tr><td>CO4</td><td>Analyze professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.</td></tr>
<tr><td>CO5</td><td>Design professional elective: cyber security concepts in engineering situations with appropriate evidence, constraints, and validation criteria.</td></tr>
</table>

<table>
<tr><th>Module No.</th><th>Unit No.</th><th>Topics</th><th>Ref.</th><th>Hrs.</th></tr>
<tr><td rowspan="5">1</td><td>1.1</td><td>Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.</td><td>--</td><td rowspan="5">9</td></tr>
<tr><td>1.2</td><td>Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.</td><td>--</td></tr>
<tr><td>1.3</td><td>Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.</td><td>--</td></tr>
<tr><td>1.4</td><td>Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.</td><td>--</td></tr>
<tr><td>1.5</td><td>Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.</td><td>--</td></tr>
<tr><td colspan="3">Total</td><td colspan="2">46</td></tr>
</table>

Module No.	Unit No.	Topics	Ref.	Hrs.
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

Recommended Books:

A. Kulkarni, M. Rao. Professional Elective: Cyber Security: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Professional Elective: Cyber Security. Pearson, 1st, 2024.
Department Faculty Board. Professional Elective: Cyber Security Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content (includes Practical)

<table>
<tr><td rowspan="3">Course Code
CSO405</td><td rowspan="3">Course Name
Open Elective: Data Visualization</td><td colspan="4">Teaching Scheme (Hrs/week)</td><td colspan="4">Credits Assigned</td></tr>
<tr><td>L</td><td>T</td><td>P</td><td>SL</td><td>L</td><td>T</td><td>P</td><td>Total</td></tr>
<tr><td>2</td><td>--</td><td>2</td><td>--</td><td>2</td><td>--</td><td>1</td><td>3</td></tr>
</table>

Examination Scheme						
	ISE1	**MSE**	**ISE2**	**ESE**	**Total**	
Theory	20	--	20	60	100	
Lab	25	--	25	--	50	

Pre-requisite Course Codes	Nil	
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design open elective: data visualization concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	**1.1**	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Module No.	Unit No.	Topics	Ref.	Hrs.
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	8
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	9
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	10
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**46**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
End Semester Examination (60 marks): Written examination based on the approved syllabus and assessment pattern.

To be Taught in laboratory			
	Topics	**Ref.**	**COs**
1	Open Elective: Data Visualization Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Open Elective: Data Visualization Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Open Elective: Data Visualization Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Open Elective: Data Visualization Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Open Elective: Data Visualization Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Open Elective: Data Visualization Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Open Elective: Data Visualization: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Open Elective: Data Visualization. Pearson, 1st, 2024.
Department Faculty Board. Open Elective: Data Visualization Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

Society of St. Francis Xavier, Pilar's

Fr. Conceicao Rodrigues College of Engineering

Fr. Agnel Ashram, Bandstand, Bandra (W), Mumbai - 400 050

(Autonomous College affiliated to University of Mumbai)

Course Content (includes Practical)

Course Code	Course Name	Teaching Scheme (Hrs/week)				Credits Assigned			
		L	T	P	SL	L	T	P	Total
CSP406	Community Innovation Project	--	--	4	--	--	--	2	2

Examination Scheme						
	ISE1	MSE	ISE2	ESE	Total	
Theory	20	--	20	--	50	
Lab	25	--	25	--	50	

Pre-requisite Course Codes	Engineering mathematics, programming fundamentals, and discipline-specific foundation courses.	
After the successful completion students should be able to:		
Course Outcomes	**CO1**	Identify community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO2	Explain community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO3	Apply community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO4	Analyze community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.
	CO5	Design community innovation project concepts in engineering situations with appropriate evidence, constraints, and validation criteria.

Module No.	Unit No.	Topics	Ref.	Hrs.
1	1.1	Foundations 1.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	1.2	Methods 1.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	1.3	Design 1.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	1.4	Analysis 1.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	1.5	Applications 1.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
2	2.1	Foundations 2.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	2.2	Methods 2.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	2.3	Design 2.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	2.4	Analysis 2.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	2.5	Applications 2.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
3	3.1	Foundations 3.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	3.2	Methods 3.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	3.3	Design 3.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	3.4	Analysis 3.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	3.5	Applications 3.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
4	4.1	Foundations 4.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	4.2	Methods 4.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	4.3	Design 4.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	4.4	Analysis 4.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	4.5	Applications 4.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**0**	

Module No.	Unit No.	Topics	Ref.	Hrs.
5	5.1	Foundations 5.1: Definitions, motivation, engineering context, notation, and representative applications.	--	0
	5.2	Methods 5.2: Core methods, implementation strategy, algorithmic trade-offs, and validation constraints.	--	
	5.3	Design 5.3: Structured design process, modelling choices, edge cases, and documentation expectations.	--	
	5.4	Analysis 5.4: Performance, correctness, complexity, maintainability, and professional engineering judgement.	--	
	5.5	Applications 5.5: Laboratory or industry examples mapped to course outcomes and assessment rubrics.	--	
Total			**0**	

Course Assessment:

Theory:

ISE-1 (20 marks): Self-learning, quiz, or formative classroom assessment.
ISE-2 (20 marks): Application-oriented internal assessment mapped to course outcomes.
Laboratory Rubric (25 marks): Continuous laboratory evaluation using predefined rubrics and viva voce.

To be Taught in laboratory			
	Topics	**Ref.**	**COs**
1	Community Innovation Project Experiment 1: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
2	Community Innovation Project Experiment 2: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
3	Community Innovation Project Experiment 3: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
4	Community Innovation Project Experiment 4: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
5	Community Innovation Project Experiment 5: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--
6	Community Innovation Project Experiment 6: Design, implement, test, document, and orally defend the assigned engineering task using approved rubrics.	--	--

Recommended Books:

A. Kulkarni, M. Rao. Community Innovation Project: Principles and Practice. University Press, 2nd, 2025.
S. Mehta. Advanced Topics in Community Innovation Project. Pearson, 1st, 2024.
Department Faculty Board. Community Innovation Project Laboratory Manual. Internal Publication, 2026, 2026.

CO-PO Mapping Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		1	2	3		1	2	3		1	2	3
CO2	1	2	3		1	2	3		1	2	3	
CO3	2	3		1	2	3		1	2	3		1
CO4	3		1	2	3		1	2	3		1	2
CO5		1	2	3		1	2	3		1	2	3

ANNEXURE A: ACADEMIC POLICIES

1. Evaluation Framework

The academic evaluation uses Continuous Internal Evaluation and End Semester Examination components. Students must secure the minimum passing threshold as defined by autonomous college regulations.

2. Grading System

Under the autonomous framework, grading follows a standard relative system converted to a 10-point CGPA scale.

3. Syllabus Revision Protocol

Major syllabus revisions occur periodically to absorb technological shifts. Minor corrections and elective introductions may occur annually with Board of Studies recommendations and Academic Council approval.

REVIEWER COMMENTS SECTION

Course	Section	Comment	Status
CSC301	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC301	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSL301	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSL301	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC302	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC302	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSL302	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSL302	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC303	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC303	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC304	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSC304	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSE305	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSE305	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSP306	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open

Course	Section	Comment	Status
CSP306	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC401	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC401	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSL401	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSL401	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC402	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSC402	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSC403	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSC403	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSE404	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSE404	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSO405	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Open
CSO405	Modules	Check whether long module blocks remain intact across print page boundaries.	Open
CSP406	Course Outcomes	Confirm outcome verbs are measurable and match assessment instruments.	Resolved
CSP406	Modules	Check whether long module blocks remain intact across print page boundaries.	Open